

WESTERN CAROLINA, NC, USA

WMO: 722177

Lat: 35.195N

Lon: 83.865W

Elev: 517

StdP: 95.27

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 63811

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.0	(c) -7.2	(d) -16.3	(e) 1.0	(f) -2.2	(g) -13.6	(h) 1.2	(i) -2.8	(j) 8.3	(k) 7.3	(l) 7.3	(m) 6.3	(n) 0.9	(o) 80	(p) 0.374

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	32.4	23.0	31.3	22.7	30.2	22.3	25.0	29.7	24.2	29.1	23.6	28.2	1.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	19.3	27.0	22.7	18.6	26.5	22.3	18.1	26.1	79.2	29.8	75.9	29.3	73.5	28.1	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.7	5.4	4.7	-13.5	33.6	3.1	1.8	-15.7	34.9	-17.5	35.9	-19.2	36.9	-21.5	38.3
			-14.0	26.0	3.1	1.1	-16.2	26.8	-18.0	27.5	-19.8	28.1	-22.0	28.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	3.8	5.6	9.7	14.3	19.0	22.5	23.8	23.4	20.7	14.6	9.0	5.8
	DBStd	8.17	5.57	5.05	4.73	4.07	3.47	2.21	1.99	2.00	2.74	4.35	4.42	4.98
	HDD10.0	647	206	138	64	10	1	0	0	0	0	11	68	149
	HDD18.3	2050	451	357	269	128	36	2	0	0	12	128	280	388
	CDD10.0	2254	13	15	54	141	279	375	427	415	322	155	39	19
	CDD18.3	615	0	0	1	8	55	127	169	157	84	14	1	0
	CDH23.3	5403	0	1	28	189	587	1102	1331	1272	719	168	6	0
	CDH26.7	1744	0	0	2	26	158	387	479	451	212	28	0	0
Wind	WSAvg	1.1	1.4	1.5	1.6	1.7	1.3	0.9	0.7	0.6	0.6	0.9	1.1	1.1
Precipitation	PrecAvg	1656	170	164	168	129	126	148	148	121	110	87	122	174
	PrecMax	2099	340	287	283	247	297	293	235	233	209	196	214	277
	PrecMin	1116	92	77	85	59	40	56	67	37	0	4	47	75
	PrecStd	238	63	51	58	47	54	58	46	52	55	54	47	63
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.4	22.1	26.3	28.9	32.2	33.7	33.7	33.0	32.1	29.3	24.0	20.7
		MCWB	14.7	15.3	15.1	16.2	21.3	22.6	23.8	23.6	21.8	20.6	15.5	15.9
	2%	DB	17.4	19.2	23.0	27.4	30.1	32.1	32.3	32.2	30.6	27.2	21.9	18.0
		MCWB	13.3	13.8	14.8	16.2	19.9	22.3	23.8	23.3	21.1	18.5	13.9	14.6
	5%	DB	15.4	17.4	21.1	25.9	28.4	30.7	31.1	31.0	29.0	25.2	19.7	16.4
		MCWB	12.3	11.9	13.7	16.2	18.8	22.0	23.7	23.3	20.9	17.3	14.2	13.4
	10%	DB	12.9	14.9	18.6	23.7	27.1	28.9	29.6	29.2	27.7	22.9	17.5	14.2
		MCWB	10.7	10.5	12.6	15.4	18.7	21.4	23.2	22.7	20.2	16.6	13.0	11.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.9	17.2	17.9	20.1	23.7	24.7	26.6	25.7	23.6	22.3	19.1	17.6
		MCDB	18.0	20.2	22.5	25.0	29.4	30.8	30.9	30.1	28.1	27.2	21.6	19.2
	2%	WB	14.5	15.5	16.4	18.9	21.8	23.9	25.2	24.7	22.9	20.6	17.0	15.8
		MCDB	16.6	18.5	20.7	23.9	26.3	29.3	29.8	29.5	27.6	23.8	19.2	17.4
	5%	WB	13.0	13.5	15.2	17.9	20.9	23.1	24.4	24.1	22.2	19.4	15.6	14.2
		MCDB	14.8	15.9	19.2	22.2	25.8	28.2	29.1	29.0	26.6	22.9	17.9	15.8
	10%	WB	11.4	11.5	13.8	16.8	20.1	22.3	23.8	23.5	21.4	18.1	14.0	12.5
		MCDB	12.8	13.6	17.3	21.1	24.9	27.1	28.1	27.9	25.5	21.5	16.4	13.8
Mean Daily Temperature Range	5% DB	MDBR	11.8	12.5	13.3	14.8	13.8	12.6	11.3	12.1	13.1	14.9	14.5	11.7
		MCDBR	13.4	15.3	17.0	18.2	16.6	14.7	13.2	13.8	15.4	16.9	17.2	13.4
		MCWBR	9.4	9.3	9.1	8.7	7.4	6.4	6.3	6.5	7.0	8.8	10.2	9.4
	5% WB	MCDBR	10.6	11.8	13.4	13.5	13.4	12.6	11.8	12.0	12.0	12.6	11.6	10.6
		MCWBR	8.3	8.7	8.5	7.8	6.8	6.1	6.1	6.1	5.9	7.1	8.6	8.5
Clear-Sky Solar Irradiance	taub		0.304	0.312	0.344	0.385	0.439	0.468	0.505	0.511	0.405	0.363	0.326	0.303
	taud		2.528	2.499	2.436	2.321	2.186	2.136	2.061	2.024	2.303	2.421	2.485	2.552
	Ebn at Noon		906	935	927	900	851	823	789	776	851	863	868	885
	Edh at Noon		81	93	108	127	147	155	166	169	122	99	84	75
All-Sky Solar Radiation	RadAvg		2.47	3.02	4.16	5.25	5.80	6.11	5.80	5.41	4.67	3.83	2.82	2.12
	RadStd		0.18	0.37	0.34	0.36	0.41	0.44	0.41	0.28	0.42	0.44	0.30	0.21

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (34 neighbors)	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+146	+96

Nomenclature: See separate page